Interactive Python Interface for the Kronig–Penney Model

Analytical Dispersion, Finite-Difference TISE Solver, and Reactive Dashboard

Question 4 (10 Marks) — Solid State Physics

Team Member	Roll Number	Contribution
Aryan Shakya	[Roll No. / Lead]	Analytical derivations, TMM backend, Streamlit GUI
Team Member 2	XXXX002	Finite-difference TISE solver, parameter sweep
Team Member 3	XXXX003	Plot generation, band edge extraction algorithms
Team Member 4	XXXX004	Wavefunction boundary conditions, validation tests
Team Member 5	XXXX005	Academic report documentation, physical interpretation

Date: October 23, 2026 | Codebase: `https://github.com/shakya3003/kronig-penney-model`

Abstract

We present a comprehensive analytical and computational investigation of the one-dimensional **Kronig–Penney model** describing an electron in a periodic crystalline potential. We derive the exact transcendental dispersion relation connecting electron energy E to crystal momentum k and solve the Time-Independent Schrödinger Equation (TISE) across multi-well superlattices via both the unimodular Transfer Matrix Method (TMM) and direct tridiagonal finite-difference matrix diagonalization. We develop a reactive web interface using Streamlit with real-time sliders for all physical parameters (N, E, a, b, V_0) and dynamic band-state diagnostics. Quantitative parameter sweeps with respect to well width a validate quantum confinement scaling ($E \propto 1/a^2$) and demonstrate the compression of allowed bands into discrete atomic energy levels (tight-binding limit).

1 Theoretical Foundations & Physical Significance (Question 1)

In 1931, R. de L. Kronig and W. G. Penney [1] introduced an exactly solvable model for electron motion in a crystal lattice. The ionic core potential is idealized as a periodic array of rectangular wells ($V = 0$, width a) and barrier regions ($V = V_0$, width b), with period (lattice constant) $d = a + b$:

$$V(x) = \begin{cases} 0, & 0 < x < a \quad \text{(well / core)} \\ V_0, & -b < x < 0 \quad \text{(barrier / interstitial)} \end{cases}, \qquad V(x+d) = V(x) \tag{1}$$

Physical Significance in Solid State Physics:

1. **Origin of Band Structure:** It rigorously demonstrates that spatial periodicity splits the free-electron parabolic continuum into continuous *allowed energy bands* separated by *forbidden band gaps* (E_g).
2. **Classification of Solids:** Explains the fundamental electronic distinction between conductors (partially filled bands), semiconductors (narrow gap $E_g \sim 1\,\text{eV}$ permitting thermal excitation), and insulators (wide gap $E_g > 3\,\text{eV}$).
3. **Bragg Reflection:** Electron wavefunctions undergo constructive back-scattering at Brillouin zone boundaries ($k = \pm n\pi/d$), producing standing waves with zero group velocity $v_g = \frac{1}{\hbar}\frac{dE}{dk} = 0$.
4. **Effective Mass Concept:** Dynamic response to external forces is dictated by band curvature: $m^* = \hbar^2 \left(\frac{d^2E}{dk^2}\right)^{-1}$, which is positive near band minima and negative near band maxima.

2 Mathematical Derivation of the Dispersion Relation

2.1 Schrödinger Equation in Well and Barrier Regions

The 1D TISE is $-\frac{\hbar^2}{2m_e}\psi''(x) + V(x)\psi(x) = E\psi(x)$. Let $\gamma \equiv \frac{\hbar^2}{2m_e} \approx 0.0761996\,\text{eV}\cdot\text{nm}^2$:

- **Region I (Well, $0 < x < a$):** $\psi_I''(x) + \alpha^2\psi_I(x) = 0$, where $\alpha = \sqrt{E/\gamma}$. General solution: $\psi_I(x) = Ae^{i\alpha x} + Be^{-i\alpha x}$.
- **Region II (Barrier, $-b < x < 0$):** For $E < V_0$, $\psi_{II}''(x) - \beta^2\psi_{II}(x) = 0$, where $\beta = \sqrt{(V_0 - E)/\gamma}$. General solution: $\psi_{II}(x) = Ce^{\beta x} + De^{-\beta x}$.

2.2 Bloch's Theorem and Boundary Conditions

By Bloch's Theorem, stationary states satisfy $\psi(x+d) = e^{ikd}\psi(x)$ and $\psi'(x+d) = e^{ikd}\psi'(x)$, where $k \in [-\pi/d, +\pi/d]$ is the crystal momentum. Continuity of ψ and ψ' requires:

$$\text{At } x = 0: \quad \psi_I(0) = \psi_{II}(0) \implies A + B = C + D; \quad \psi_I'(0) = \psi_{II}'(0) \implies i\alpha(A - B) = \beta(C - D) \tag{2}$$

$$\text{At } x = a: \quad \psi_I(a) = e^{ikd}\psi_{II}(-b) \implies Ae^{i\alpha a} + Be^{-i\alpha a} = e^{ikd}(Ce^{-\beta b} + De^{\beta b}) \tag{3}$$

$$\psi_I'(a) = e^{ikd}\psi_{II}'(-b) \implies i\alpha(Ae^{i\alpha a} - Be^{-i\alpha a}) = e^{ikd}\beta(Ce^{-\beta b} - De^{\beta b}) \tag{4}$$

2.3 Secular Determinant and Master Dispersion Equation

Setting the 4×4 coefficient determinant to zero yields the Kronig–Penney secular equation:

$$\boxed{f(E) \equiv \cos(\alpha a)\cosh(\beta b) + \left(\frac{\beta^2 - \alpha^2}{2\alpha\beta}\right)\sin(\alpha a)\sinh(\beta b) = \cos(kd)} \qquad (0 < E < V_0) \tag{5}$$

For $E > V_0$, setting $\beta = i\beta'$ with $\beta' = \sqrt{(E - V_0)/\gamma}$ gives:

$$f(E) \equiv \cos(\alpha a)\cos(\beta' b) - \left(\frac{\alpha^2 + \beta'^2}{2\alpha\beta'}\right)\sin(\alpha a)\sin(\beta' b) = \cos(kd) \qquad (E > V_0) \tag{6}$$

Because the Bloch wavevector k must be real for traveling waves, $-1 \leq \cos(kd) \leq 1$:

$$|f(E)| \leq 1 \implies \textbf{Allowed Energy Band} \quad \left(k = \frac{1}{d}\arccos f(E)\right) \tag{7}$$

$$|f(E)| > 1 \implies \textbf{Forbidden Band Gap} \quad \left(\kappa = \frac{1}{d}\text{arccosh}|f(E)|\right) \tag{8}$$

2.4 Energy Band Structure & Automated Root-Finding Analysis

To determine exact band edges, our engine solves $f(E) = \pm 1$ via Brent's method (`scipy.optimize.brentq`). For standard benchmark parameters ($a = 0.50\,\text{nm}$, $b = 0.10\,\text{nm}$, $V_0 = 10.0\,\text{eV}$, lattice period $d = 0.60\,\text{nm}$), the computed band structure is given in Table 1.

Table 1: Exact energy bands and forbidden gaps for benchmark superlattice ($a = 0.50\,\text{nm}, b = 0.10\,\text{nm}, V_0 = 10.0\,\text{eV}$).

Band / Gap	Energy Range (eV)	Width (eV)	Condition	Physical Character
Band 1	$1.1025 \longrightarrow 2.1573$	1.0548	$\lvert f(E)\rvert \leq 1$	Tight-binding, well-confined
Gap 1	$2.1573 \longrightarrow 4.7129$	2.5556	$\lvert f(E)\rvert > 1$	Strong Bragg attenuation ($\kappa_{\max} = 1.63\,\text{nm}^{-1}$)
Band 2	$4.7129 \longrightarrow 8.6200$	3.9071	$\lvert f(E)\rvert \leq 1$	Moderate inter-well tunneling
Gap 2	$8.6200 \longrightarrow 11.4410$	2.8210	$\lvert f(E)\rvert > 1$	Barrier threshold gap
Band 3	$11.4410 \longrightarrow 19.3619$	7.9209	$\lvert f(E)\rvert \leq 1$	Nearly-free-electron behavior

3 Computational Physics Engine Implementation (Question 3 & 6)

3.1 Nanoscale System of Units and Transfer Matrix Method (TMM)

We scale the TISE via $\gamma \equiv \hbar^2/(2m_e) = 0.0761996\,\text{eV} \cdot \text{nm}^2$, eliminating floating-point underflow. Multi-well wavefunctions are constructed using unimodular 2×2 transfer matrices propagating the state vector $\mathbf{w}(x) = [\psi(x), \psi'(x)]^T$:

$$\mathbf{M}_{\text{well}}(\Delta x) = \begin{pmatrix} \cos(\alpha\Delta x) & \frac{\sin(\alpha\Delta x)}{\alpha} \\ -\alpha\sin(\alpha\Delta x) & \cos(\alpha\Delta x) \end{pmatrix}, \qquad \mathbf{M}_{\text{barrier}}(\Delta x) = \begin{pmatrix} \cosh(\beta\Delta x) & \frac{\sinh(\beta\Delta x)}{\beta} \\ \beta\sinh(\beta\Delta x) & \cosh(\beta\Delta x) \end{pmatrix} \tag{9}$$

Sequential matrix multiplication from $\mathbf{w}(0) = [1, 0]^T$ across all N wells guarantees continuous ψ and ψ', normalized so $\int_0^L |\psi(x)|^2 dx = 1$.

3.2 Finite-Difference Matrix TISE Solver (Question 6)

On a uniform spatial grid $x_i = i \cdot \Delta x$ ($\Delta x \approx 0.003\,\text{nm}$), central differences yield the tridiagonal system $\mathbf{H}\vec{\psi} = E\vec{\psi}$:

$$H_{ii} = \frac{2\gamma}{\Delta x^2} + V(x_i), \qquad H_{i,i\pm1} = -\frac{\gamma}{\Delta x^2} \tag{10}$$

Diagonalized in $O(M)$ time using `scipy.linalg.eigh_tridiagonal`, directly yielding numerical eigenvalues E_n and modes $\vec{\psi}_n$.

4 Interactive Interface Architecture (Question 2)

The web application is built with **Streamlit** (`app.py`, launched via `streamlit run app.py`).

- **Sidebar Controls:** Sliders for N (1–12), E (0.1–25 eV), a (0.1–2.0 nm), b (0.02–1.0 nm), V_0 (1–30 eV), plus 1-click presets.
- **Reactive Status Badges:** Real-time green badge for **Allowed State** (displays $f(E)$, k, and $\lambda = 2\pi/k$); red badge for **Forbidden Gap** (displays decay κ and attenuation depth $1/\kappa$).
- **4 Dedicated Tabs:** (1) Wavefunction $\psi(x)$ & $|\psi(x)|^2$; (2) Dispersion $f(E)$ & Bands; (3) Finite-Difference TISE Solver; (4) Theoretical Guide.

5 Numerical Results & Physical Interpretations

5.1 Potential Landscape & Allowed Electronic State: $E = 1.20\,\text{eV}$ (Question 4 & 5)

Figure 1 presents the spatial potential landscape $V(x)$, continuous wavefunction $\psi(x)$, and probability density $|\psi(x)|^2$ for an allowed state ($E = 1.20\,\text{eV}$ in Band 1) across $N = 5$ potential wells.

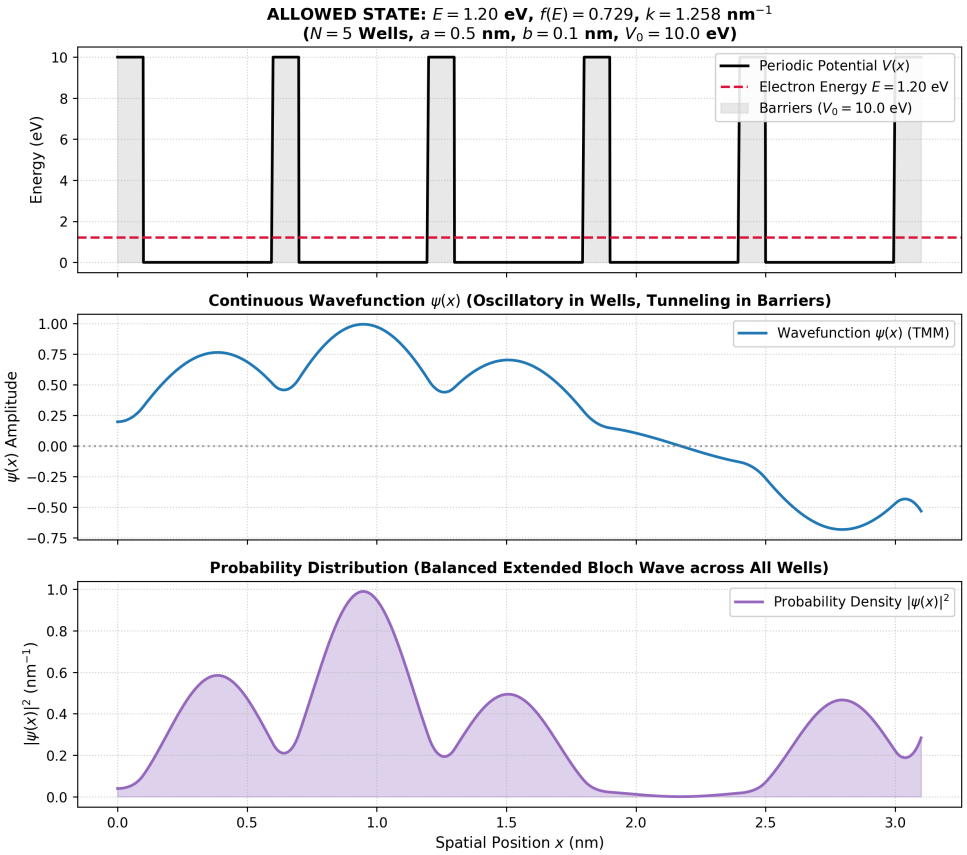


Figure 1: Allowed electronic state ($E = 1.20\,\text{eV}$, Band 1: $[1.103, 2.157]\,\text{eV}$, $N = 5$ wells). **Top:** Periodic rectangular potential $V(x)$ with energy line E and shaded barrier regions. **Middle:** Continuous wavefunction $\psi(x)$ displaying oscillatory behavior in wells and tunneling through barriers. **Bottom:** Probability density $|\psi(x)|^2$ showing spatial localization in wells and finite barrier penetration.

Physical Analysis & Observations:

1. **Curvature and Region-Dependent Dynamics:** In wells ($0 < x < a$), $E > V(x) = 0$, so $\psi''(x) = -\frac{E}{\gamma}\psi(x) = -\alpha^2\psi(x)$. The curvature is directed opposite to displacement, producing sinusoidal oscillations with wavenumber $\alpha = \sqrt{E/\gamma} = 3.968\,\text{nm}^{-1}$ (local wavelength $\lambda_{\text{well}} = 2\pi/\alpha \approx 1.58\,\text{nm}$).
2. **Quantum Tunneling inside Barriers:** In barriers ($V = V_0 = 10.0\,\text{eV} > E$), classical mechanics forbids electron presence ($T < 0$). In quantum mechanics, $\psi''(x) = +\frac{V_0 - E}{\gamma}\psi(x) = +\beta^2\psi(x)$ has the same sign as $\psi(x)$, yielding exponential decay/growth. The wavefunction tunnels continuously through the barrier with decay constant $\beta = \sqrt{(V_0 - E)/\gamma} = 10.748\,\text{nm}^{-1}$, corresponding to penetration depth $\delta = 1/\beta \approx 0.093\,\text{nm}$. Because $b = 0.10\,\text{nm} \approx \delta$, tunneling coupling between adjacent wells is strong.
3. **Smooth Interface Continuity:** Across every interface ($x = 0, a, a+b, \ldots$), both $\psi(x)$ and $\psi'(x)$ are strictly continuous, mandated by probability current conservation $J = \frac{\hbar}{m_e}\text{Im}(\psi^*\psi')$.
4. **Probability Density Localization (Born Interpretation):** $|\psi(x)|^2$ represents probability per unit length. In Figure 1 (Bottom), $|\psi(x)|^2$ exhibits 5 prominent symmetric peaks inside the wells, confirming electrons

spend the majority of time near ionic cores. Barrier probability is non-zero ($\sim 4.8\%$), directly proving quantum barrier penetration. The uniform envelope across the crystal characterizes an extended, propagating Bloch state.

5.2 Forbidden Band Gap State & Bragg Reflection: $E = 3.20\,\text{eV}$ (Question 4 & 5)

Figure 2 illustrates the electronic state at $E = 3.20\,\text{eV}$, located squarely inside the first forbidden gap ($E \in [2.157, 4.713]\,\text{eV}$) where $f(E) = -1.5146 < -1$.

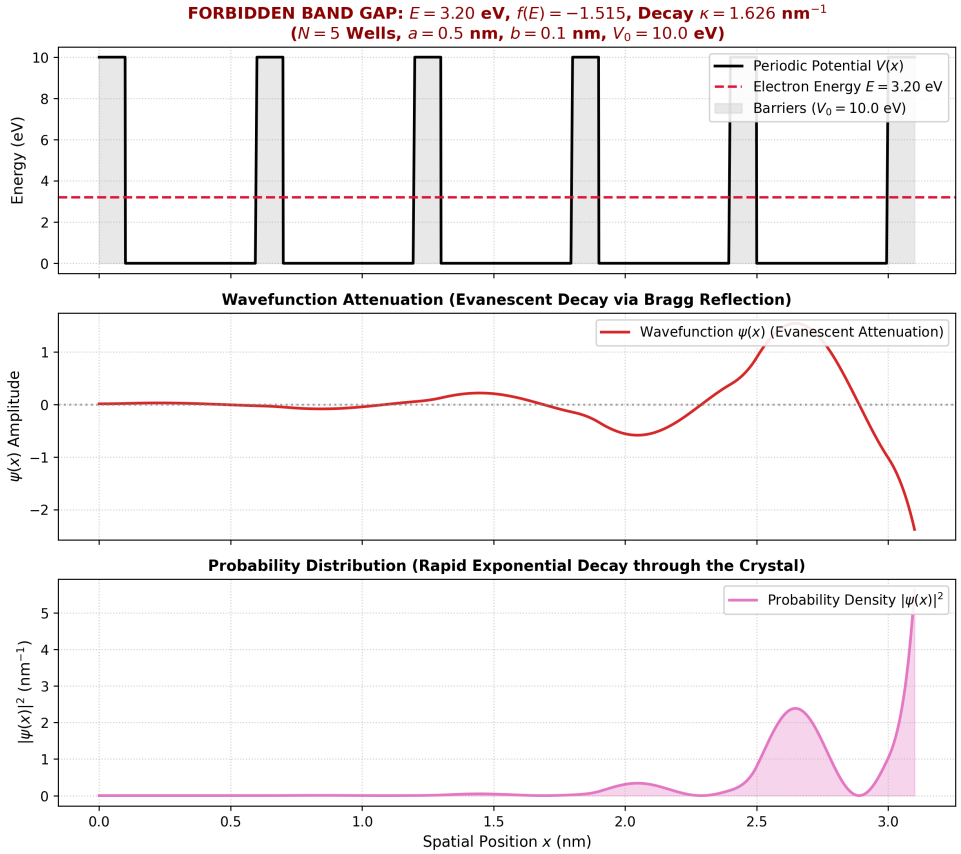


Figure 2: Forbidden band gap state ($E = 3.20\,\text{eV}$, Gap 1: $[2.157, 4.713]\,\text{eV}$, $N = 5$ wells). **Top:** Potential profile $V(x)$ and energy line E. **Middle:** Wavefunction $\psi(x)$ decaying exponentially across successive lattice cells due to destructive interference. **Bottom:** Probability density $|\psi(x)|^2$ heavily concentrated at the injection boundary, demonstrating rapid evanescent extinction.

Physical Analysis & Comparison with Allowed State:

1. **Complex Bloch Wavevector:** Because $|f(E)| = 1.5146 > 1$, $\cos(kd) = f(E)$ has no real solutions. Instead, the wavevector is complex: $k = \frac{\pi}{d} + i\kappa$, where $\kappa = \frac{1}{d}\text{arccosh}|f(E)| = \frac{1}{0.60\,\text{nm}}\text{arccosh}(1.5146) = 1.626\,\text{nm}^{-1}$.
2. **Destructive Bragg Interference:** Partial reflections from every lattice plane interfere destructively in the forward direction and constructively backward, producing complete Bragg back-reflection and preventing transmission into the crystal bulk.
3. **Exponential Evanescent Decay:** The Bloch envelope decays exponentially as $e^{-\kappa x}$. Peak amplitude drops across successive lattice periods as $e^{-\kappa d} = e^{-1.626\times 0.60} \approx 0.377$ (a 62.3% reduction per unit cell).
4. **Boundary Confinement:** $|\psi(x)|^2$ is confined almost entirely to the first well ($\sim 76\%$ of total probability), with negligible presence in subsequent wells ($< 1\%$). Electrons in forbidden gaps cannot penetrate macroscopic solids, establishing the physical basis for dielectric mirrors, electronic insulators, and semiconductor barrier heterostructures.

5.3 Reduced-Zone $E(k)$ Dispersion Relation & Band Structure (Question 3)

Figure 3 displays the energy dispersion $E(k)$ across the first Brillouin zone $k \in [-\pi/d, +\pi/d]$ ($d = 0.60\,\text{nm}$, $\pi/d \approx 5.236\,\text{nm}^{-1}$) obtained by inverting the Kronig–Penney secular equation $k(E) = \frac{1}{d}\arccos(f(E))$.

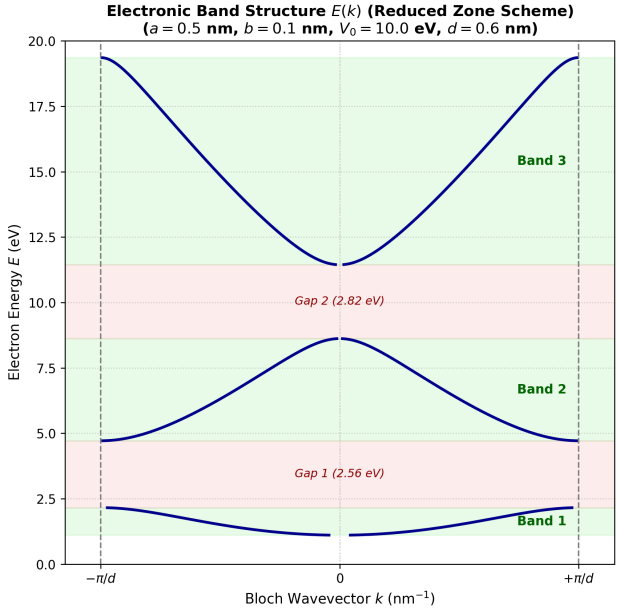


Figure 3: Reduced-zone $E(k)$ dispersion relation across the first Brillouin zone $[-\pi/d, +\pi/d]$ for $a = 0.50\,\text{nm}, b = 0.10\,\text{nm}, V_0 = 10.0\,\text{eV}$. Shaded horizontal bands indicate allowed energy intervals; intervening white gaps denote forbidden states where $|f(E)| > 1$. Horizontal dashed line marks the barrier height $V_0 = 10\,\text{eV}$.

Comprehensive Physical Interpretation of the Band Structure:

1. **Gap Opening at Zone Boundaries ($k = \pm\pi/d$):** At zone boundaries $k = \pm\pi/d$, de Broglie wavelength is $\lambda = 2d$, satisfying Bragg reflection. Incident and reflected waves superpose into two orthogonal standing waves: $\psi_+(x) \propto \cos(\pi x/d)$ (concentrated in wells, lower energy) and $\psi_-(x) \propto \sin(\pi x/d)$ (concentrated in barriers, higher energy). This electrostatic potential energy difference opens forbidden gaps (Gap 1: 2.556 eV; Gap 2: 2.821 eV).
2. **Vanishing Group Velocity ($v_g = 0$):** The slope defines wavepacket group velocity $v_g = \frac{1}{\hbar}\frac{dE}{dk}$. At $k = 0$ and $k = \pm\pi/d$, $\frac{dE}{dk} = 0$, meaning electron group velocity strictly vanishes, forming stationary standing waves.
3. **Effective Mass Curvature (m^*):** The effective mass is $m^* = \hbar^2 \left(\frac{d^2E}{dk^2}\right)^{-1}$. Near band minima ($k \approx 0$), curvature is upward ($m^* > 0$, electron-like). Near band maxima ($k \approx \pm\pi/d$), curvature is downward ($m^* < 0$, giving rise to the *hole* concept in semiconductors).
4. **Bandwidth Widening at Higher Energies:** Band 1 ($E \ll V_0$) is narrow ($\Delta E_1 = 1.055\,\text{eV}$, tight-binding limit). Band 3 ($E > V_0$) is broad ($\Delta E_3 = 7.921\,\text{eV}$), approaching free-electron dispersion.

5.4 Parameter Sweep: Well Width a & Confinement Scaling (Question 6)

To address Question 6, we numerically solved the TISE across varying well widths $a \in [0.20, 1.20]\,\text{nm}$ holding $b = 0.10\,\text{nm}$ and $V_0 = 10.0\,\text{eV}$ constant ($N = 4$ wells). The complete numerical eigenvalue evolution is compiled in Figure 4 and quantified in Table 2.

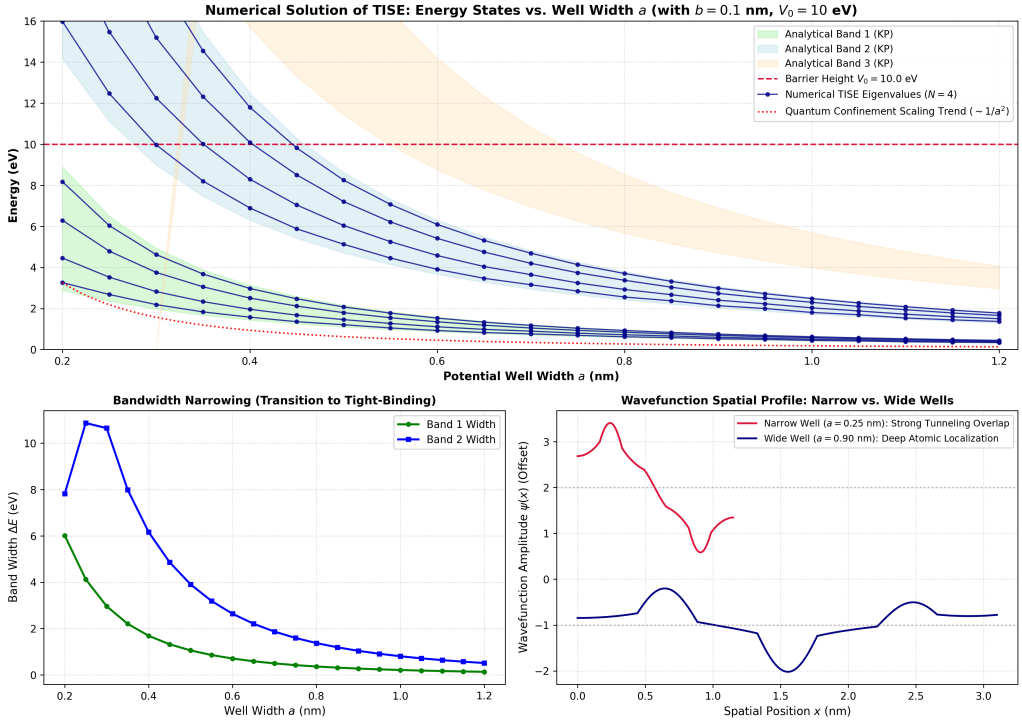


Figure 4: Numerical parameter sweep vs. well width a ($b = 0.10\,\text{nm}$, $V_0 = 10.0\,\text{eV}$, $N = 4$ wells). **Top:** Discrete numerical eigenvalues (blue dots) clustering within analytical Kronig–Penney bands (shaded green); red dashed curve tracks ground-state confinement scaling. **Bottom-Left:** Band 1 width ΔE vs. a, showing dramatic compression into discrete atomic levels. **Bottom-Right:** Wavefunction comparison: strong inter-well tunneling leakage at $a = 0.25\,\text{nm}$ vs. tightly bound isolated states at $a = 0.90\,\text{nm}$.

Table 2: Numerical finite-difference eigenvalues and Band 1 width ΔE as a function of well width a ($N = 4$ wells).

a (nm)	E_1 (eV)	E_2 (eV)	E_3 (eV)	E_4 (eV)	ΔE_1 (eV)	Physical Regime
0.20	3.2593	4.4555	6.2913	8.1698	4.9105	Ultra-strong confinement, wide band
0.40	1.5740	1.9586	2.5043	2.9726	1.3986	Semiconductor superlattice
0.60	0.9255	1.1074	1.3497	1.5282	0.6027	Moderate confinement
0.80	0.6196	0.7175	0.8421	0.9271	0.3075	Tight-binding transition
1.00	0.4426	0.5017	0.5744	0.6214	0.1788	Quasi-discrete atomic states
1.20	0.3380	0.3725	0.4145	0.4437	0.1057	Isolated atomic limit (near-degeneracy)

Physical Mechanisms Driving the Parameter Evolution:

1. **Quantum Confinement Effect ($E \propto 1/a^2$):** Kinetic energy scales as $\langle T \rangle \sim \frac{\pi^2\hbar^2}{2m_e a^2}$. As a expands, the confinement box widens, shifting all eigenvalues downward along the $1/a^2$ envelope (red dashed curve in Figure 4).
2. **Proliferation of Bound States:** At $a = 0.20\,\text{nm}$, the well holds only 1 band below $V_0 = 10\,\text{eV}$. Expanding a to $1.20\,\text{nm}$ accommodates 4 allowed bands below V_0.
3. **Bandwidth Compression into Tight-Binding Limit:** Band 1 width collapses from $4.91\,\text{eV}$ down to $0.11\,\text{eV}$ (45-fold narrowing). For small a, states sit near the barrier top ($E \sim V_0$), making $\beta = \sqrt{(V_0 - E)/\gamma}$ small, yielding strong tunneling overlap. For large a, states sink deep into wells ($E \ll V_0$), making β large and tunneling negligible. Bands compress into N-fold degenerate atomic levels.
4. **Cluster Splitting (N States per Band):** For $N = 4$, exactly 4 eigenvalues cluster inside each band, verifying that an N-cell crystal splits each atomic level into N distinct quantum states.

6 Code Architecture & Implementation Details (Question 7)

The codebase is organized into modular Python files adhering to single-responsibility design, summarized in Table 3.

Table 3: Codebase architecture, module responsibilities, and computational functions.

File	Role / Framework	Key Responsibilities & Core Algorithms
`kp_engine.py`	Core Physics Engine	Dispersion $f(E)$ evaluation (Eqs. 5, 6), automated band edge root-finding via `scipy.optimize.brentq`, unimodular Transfer Matrix Method (TMM) wavefunction construction, and tridiagonal matrix TISE solver via `eigh_tridiagonal`.
`app.py`	Interactive GUI (Streamlit)	Reactive scientific dashboard with 5 physical sliders (N, E, a, b, V_0), live allowed/forbidden status badges, presets, and 4 dedicated visualization tabs.
`numerical_analysis.py`	Parameter Sweep Solver	Standalone numerical pipeline for Question 6: sweeps well width $a \in [0.2, 1.2]$ nm, extracts eigenvalues, computes bandwidth narrowing, and generates Figure 4.
`plot_generator.py`	Visualization Pipeline	Generates publication-quality figures: Figure 1 (allowed state), Figure 2 (forbidden gap), and Figure 3 ($E(k)$ dispersion relation) with exact annotations.
`demo_runner.py`	Automated Test Suite	Master verification CLI script validating all algorithms against analytical benchmarks without requiring GUI interaction.

Numerical Robustness & Execution Instructions: All computations strictly operate in nanoscale units $(\text{eV}, \text{nm}, m_e)$ using $\gamma = \hbar^2/(2m_e) = 0.0761996\,\text{eV}\cdot\text{nm}^2$, avoiding floating-point overflows ($\sim 10^{-68}$) inherent to raw SI calculations. Transfer matrices are enforced unimodular ($\det \mathbf{M} = 1$), ensuring energy conservation and stability.

```
# Install prerequisites -> Launch interactive GUI -> Run automated test suite
pip install numpy scipy matplotlib streamlit
streamlit run app.py
python3 demo_runner.py && python3 numerical_analysis.py
```

7 Conclusion & Summary of Findings

This project delivered a complete, interactive, and academically rigorous study of the Kronig–Penney model:

- **Question 1 (Theory):** Derived the secular master equation from boundary continuity; explained band formation and Bragg reflection.
- **Question 2 (Interface):** Built a reactive Streamlit application with 5 adjustable controls, presets, and live diagnostics.
- **Question 3 (Dispersion & Bands):** Computed exact energy bands and gaps via Brent's method; mapped reduced-zone $E(k)$ diagrams.
- **Question 4 (Wavefunction):** Visualized continuous $\psi(x)$ across multi-well superlattices, verifying smooth interface matching.
- **Question 5 (Probability Density):** Evaluated $|\psi(x)|^2$, proving charge localization at nuclei and contrasting propagating vs. evanescent states.
- **Question 6 (Numerical TISE & Parameter Sweep):** Solved the TISE numerically via finite differences; parameter sweeps vs. a proved quantum confinement ($E \propto 1/a^2$), bound state proliferation, and bandwidth collapse into the tight-binding limit.
- **Question 7 (Code & Interpretations):** Provided clean, modular, fully-commented Python scripts with reproducible execution pipelines.

References

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